

BULLETIN SN & SM BOW THRUSTER

Gear Misalignment – Critical Bevel Gear Mesh & End Cap Depth Tolerances

APPLICABILITY	CRITICAL COMPONENT	CORE DOMAINS
Santorin (SN) & Super Maramu (SM) Bow Thruster End Cap & Gear Assemblies	Propeller Shaft Round End Cap, Internal Bearings & Bevel Drive Gears	Gear Mesh Alignment, End Cap Depth Sizing (10mm vs 13mm), Spacer Adaptation

CRITICAL ALERT: END CAP MISMATCH CAUSES CATASTROPHIC GEAR FAILURE

The round End Cap installed on the thruster assembly sets the axial position of the shaft bearings and determines the exact mesh depth of the bevel gears. Using an unmatched or incorrect end cap will cause immediate tooth stripping, binding, and possible propeller axle fracture under operational load



1. ROOT CAUSE & DIMENSIONAL VARIANCE

The issue with misalignment of the vertical shaft gear to the propeller shaft gear is typically caused by mixing vintage end caps with newer replacement gear sets. The round end caps shipped by Amel determine the depth of engagement between the two gears, and historical factory caps have varied in inside recess depth from **10 mm to 13 mm**.

Recent parts testing demonstrated that with replacement gear sets, the internal cap depth **must be 10 mm**. Field reports confirm that not only have individual owners inadvertently mixed vintage components, but mixed part depths (such as 11.5 mm caps) have also been distributed in replacement shipments.

END CAP VARIANT	INSIDE DEPTH	GEAR MESH RESULT	REQUIRED ACTION / CORRECTION
Old Style Cap	13.0 mm	Loose mesh or axle bind depending on vintage	Insert precision Delrin spacer to reduce depth to 10.0 mm
Intermediate Cap	11.5 mm	Improper tooth contact; 1.5 mm excessive play	Machine/shim cap depth down to exact 10.0 mm specification
Correct Target Spec	10.0 mm	Correct parallel mesh and rotational alignment	Direct fit; check shaft manual rotation before sealing



2. FIELD ALIGNMENT & INSPECTION PROTOCOL

- ☐ **Visual Mesh Inspection:** Conduct multiple dry trial assemblies. Verify that gear teeth seat sufficiently deep without bottoming out or exhibiting binding tight spots
- ☐ **Manual Rotation Verification:** Manually turn the input and output shafts. They should rotate freely and smoothly by hand, with the gears meshing evenly
- ☐ **Gear Mesh Verification:** Download the companion technical guide: Amel Book Supplement – Bow Thruster Gear Mesh Check.pdf. and follow the step-by-step procedures. If you are not an Amel School client, email Bill@AmelSchool.com and request the Amel Book Supplement. Note: If not a client, the first half-hour of Amel Support & Coaching is FREE.

- ☐ **Fastener Clocking:** Cap securing screws are indexed at 90° intervals. Locate one hole precisely to properly align all remaining fasteners

3. TECHNICAL REFERENCES & FACTORY CONTACT

For detailed step-by-step backlash and tooth engagement measurement procedures, consult the companion technical guide: **Amel Book Supplement – Bow Thruster Gear Mesh Check.pdf**. **Note: Amel Customer Service was formally notified of component variance on 6 July 2023. The errors made by Amel may be solved, but if you have spare parts for this job, they may not be matched.**



To better illustrate gear mesh, I used a photo of yellow gear paste rather than Prussian Blue, or engineer's blue).



Preferred Method: Use a non-drying Gear Marking Compound (often called Prussian Blue. I used a photo of yellow gear paste rather than Prussian paste, or engineer's blue) to better illustrate gear mesh. In the photo at left, it is a specialized gear yellow paste formulation. Because you are doing a blind assembly, this method requires a very specific protocol to ensure the pattern you read after installing a new gear set is accurate. Prussian Blue is the preferred compound.

Prussian Blue data Sheet ([click here](#))

